Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/23/2010
PI Name:	Souza, Richard B. Ph.D.		
Project Title:	Effects of Abnormal Loading on Tibiofemoral Articular Cartilage Composition		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	richard.souza@radiology.ucsf.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	510-235-1978
Organization Name:	University of California, San Francisco		
PI Address 1:	185 Berry Street, Suite 350		
PI Address 2:	Radiology and Biomedical Imaging		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94107	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-03 Postdoctoral Fellowships
Start Date:	11/01/2008	**End Date:**	12/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2009, from 10/30/2010, per N. Gibbins/NSBRI (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Majumdar, Sharmila (MENTOR/University of California, San Francisco)		
Grant/Contract No.:	NCC 9-58-PF01903		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP While several studies have investigated the influence of long-duration space flight on muscle and bone tissue, it remains unclear how unloading influences cartilage health. It has been shown in animals that decreased loading results in cartilage deterioration with compositional changes similar to those of osteoarthritis (OA). However, it is unknown if these same changes are experienced in astronauts during space flight. If long-duration unloading results in biochemical changes within the cartilage, it may be a medical risk-factor for the development of OA later in life. OA is a progressive degenerative disease that is characterized by changes in proteoglycan and water content. Recent advancement in MR technology allow for an increased ability to monitor these biochemical levels in vivo. Furthermore, the reversibility reported in animal work with remobilization has yet to be observed in humans. The proposed study will utilize state-of-the-art MR techniques to monitor knee cartilage composition with non-weight-bearing (simulated zero-gravity) and return to full-weight-bearing (relative increased loading). The primary objective of this proposal is to quantify changes in cartilage composition with loading deprivation using advanced MR techniques (including T2 and T1rho relaxation time mapping). A secondary objective is to demonstrate reversibility of compositional changes of articular cartilage associated with OA in vivo. The results of this study are necessary to determine if biochemical changes associated with early OA occurs in humans after a period of non-weight-bearing and to determine how these changes respond when returning to full weight bearing. Future studies will use this information to design cartilage-protecting protocols of in-space exercise programs as well as strategies for post-flight re-integration into full-gravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study are necessary to determine if biochemical changes associated with early OA occurs in humans after a period of non-weight-bearing and to determine how these changes respond when returning to full weight bearing. Future studies will use this information to design cartilage-protecting protocols of in-space exercise programs as well as post-flight re-integration into full-gravity conditions.
Task Progress:	SUPPLEMENTAL INFORMATION FOR FINAL REPORT (May 2010): Eight subjects completed. One subject currently enrolled. Findings are approaching significance already. I will consider terminating study at n=15 if findings continue to look as robust as preliminary data suggests. ANNUAL REPORTING OCTOBER 2009: Institutional Review Approval was received at the end of 2008. In January and February 2009, piloting and protocol development was completed. Between March 2009 and September 2009, eight subjects have been enrolled in the study. One subject failed to complete the follow-up session. Three subjects have completed all data collection sessions and four additional subjects are currently in the non-weight-bearing phase. Data analyis is being performed on an on-going basis and data analysis will continue as new data sets are completed.
Bibliography Type:	Description: (Last Updated:)